# PhD Position in "Collaborative multimodal perception for automated vehicles"

## Research Proposal

Applicant: Konark Karna, MSc Advanced Computer Science, Northumbria University, UK

1. **Aim**

   To improve safety and efficiency of autonomous vehicles by collaborating real-time on perception with multimodal sensor inputs i.e, RGB, Lidar, Radar, IMU, etc.

2. **Background**

   Autonomous systems - unmanned ground vehicles (UGV), Unmanned-Aerial Vehicles (UAV) or Autonomous Vehicles / Automated Driving Systems (AV/ADS) are one of the technological epitome of this century.

   In our case of AV/ADS, single-vehicle perception systems have many downsides such as lack of long-range, occlusion, sensor malfunction and noise. Therefore, researchers have made their move on Vehicle-to-Vehicle (V2V) and Vehicle-to-Everything (V2X) systems. Vehicles share information with each other based on communication technique selected among early, intermediate or late, in V2V systems (Hu *et al.*, 2025); whereas in V2X systems, additional input is mainly obtained from the Road-Side Units (RSU). V2X is preferred for RSU being on a certain height and stationary contributes less-noisy, and clearer longer-range perception information.

   However, sensors would vary among vehicles in real-scenarios. Therefore, in an ideal situation, we would need a multi-modal collaborative system that would share multi-modal perception information among all V2X systems agents. Currently, research has been quite advanced in two fronts: 1. Collaborative Perception among V2V / V2X agents for a single modality information such as either Lidar or RGB data. 2. Multi-modal sensor fusion for a single vehicle perception.

   Collaborative perception with V2V/V2X aims to tackle occlusion and missing information head-on, but it has its own drawbacks such as, voluminous data, communication latency, and pose error (non-uniform spatial coordinates) (Hu *et al.*, 2025). One of the preliminary solutions is to use intermediate fusion, and it is a preferred method in almost all methods. Next, to tackle, pose error, a method CoBEVT uses Fused Axial Attention (FAX) to capture local and global spatial information sparsely (Xu, R *et al.*, 2022a) in V2V scenario. In V2VFormer method, researchers further proposed a Spatial Aware Transformer (SAT) to re-calibrate feature significance of nearby agents as per their spatial correlation and Channel-Wise Transformer (CWT) to further underline semantic interaction among agents (Lin, C. *et al.*, 2024). In V2X scenario, we have V2X-ViT method that uses heterogenous multi-agent self-attention (HMSA) and multi-scale window attention (MSwin) in vision transformer alongside delay-aware positional embedding to correct delay-induced global spatial misalignment (Xu R, *et al.*, 2022b). Then, we have two methods DiscoNet and Core using the early collaboration method. DiscoNet uses knowledge-distillation through the teacher-student paradigm (Li *et al.*, 2021), whereas CORE tries to reconstruct an obstructed view by learning what information is needed for an agent for a particular spatial location (Wang B *et al.,* 2023). Then for the missing information situation in the V2X scenario, we have the V2X-INOP method that uses historical information to predict what might be actually in place of misdetection at current timestep (Ren S. *et al.,* 2024). Next, for minimizing bandwidth utilization in both scenarios of V2V and V2X, we have a few methods. First one - When2comm decides when to open transmission based on sufficiency of information to make its own perception task (LinY et al., 2020). Then, we have a Where2comm method that only allows agents to share spatial sparse but critical information needed to make improved perception (Hu Y. *et al*., 2022). Lastly, we have How2comm that first uses mutual-information aware communication (MIC) mechanism to select most informative messages to reduce on space and bandwidth, then with a Flow-guided delay compensation (FDC) eliminates fusion error due to temporal dissonance.

When it comes to multi-modal sensor fusion, we have a few methods such as DeepFusion and MVFusion. DeepFusion can incorporate all three primary modalities - Lidar, RGB, and Radar. It uses 2D-CNN to extract features from RGB and PIXOR(Yang *et al.,* 2019) method to extract from lidar/radar, before transforming them into BEV representation prior to fusion. Spatial and semantic alignment is performed prior to fusion as well (Drews *et al.*, 2022). MVFusion focuses on radar-camera fusion, It proposed semantic-aligned radar encoder (SARE) to semantically align it with RGB prior to incorporating in a fusion transformer (Wu *et al.*, 2023). We also have a LXL method that incorporates newer 4d mm-wave radar with image data (Xiong *et al.,* 2023), and InterFusion for newer 4D radar and lidar point clouds (Wang *et al.*, 2022). Then, we have a lot of methods proposed for lidar-camera fusion.

## 3. Methodology

Our research will be focussed on the amalgamation of two branches discussed above: collaborative perception, and multi-modal sensor fusion. However, we will have a few aspects borrowed from earlier research work. For instance, for collaboration, it is established that intermediate fusion methods provide the best trade off between data and accuracy. So, it will be prudent to follow intermediate fusion. Next, for feature extraction from RGB data, we can use multiple methods ranging from simple 2D-CNN to SwinTransformer. For Lidar data, in most research methods discussed earlier, PointPillar(Lang *et al.,* 2019) method is used for feature extraction. Lastly, for radar, either we can follow the PIXOR method as in DeepFusion or PointPillar as in the LXL and InterFusion. Another method named SECOND (Yan, Mao and Li, 2018) can be used as well for Lidar and Radar feature extraction. We, however, can conduct further review on newer 4D mm-wave radar perception methods to incorporate the most advanced feature extraction technique.

When it comes to reducing communication bandwidth for multi-modal collaboration, it will be wise to work on intuitive-tweak within transformer architecture as in the case of How2comm method. The way V2X-ViT have tweaked the transformer to introduce HMSA and MSwin blocks, we can work on the development of similar blocks that can not only fix temporal misalignment but deduce when to communicate and where within the V2X collaboration system.

Autonomous systems modelling for collaborative perception have limited available dataset, and among them OPV2V (Xu *et al.,* 2022) and V2V4Real (Xu *et al.*, 2023) are specifically for V2V cases. We, however, can work on them to evaluate our proposed methods for V2V cases. Next, we have V2X-Sim, it is for V2X environment, however, not specifically for radar (Li *et al.*, 2022) In these two cases, we need to purposefully extract radar features with the same method we are extracting Lidar features. We only have two datasets that can be utilized with all modalities of lidar, rgb, and radar in V2X scenario i.e., nuScene(Caesar *et al.,* 2020) and V2X-radar (Yang *et al.,* 2024). V2X-radar is a real-world dataset. Therefore, it shall be deemed the benchmark on which our proposed method must perform well over other existing methods.

## 4. References


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